SCD: SPATIAL CONCEPT DISTILLATION FOR SPURIOUS-CORRELATION ROBUSTNESS IN SELF-SUPERVISED PRETRAINING

ABSTRACT

Self-supervised representations inherit the co-occurrences of their unlabeled training images: when cats appear mostly indoors, an encoder trained from scratch learns indoor scenes as evidence for cats and fails on cats outdoors. Methods that counter this usually assume which feature is spurious, such as the background or the easiest feature. We introduce Spatial Concept Distillation (SCD), which makes no such assumption and tells the encoder *where* each named concept sits in an image. Before training, a frozen vision-language teacher (CLIP with a sparse concept decomposition) names the concepts of the training set and computes a dense map of every concept for every training image. During self-supervised pretraining, every location of the student's last feature map has to predict the concepts at its place in the view. At the level of whole images an object and its usual surroundings arrive together; at the level of locations they occupy different places, so the student needs a separate code for each. The prediction runs through a linear readout fitted on one half of the batch and scored on the other half, so memorizing individual images earns nothing and only concept codes shared across images lower the loss. No class label, group annotation or concept type enters pretraining and only the student encoder is kept for inference. We evaluate SCD against SimCLR and against our earlier relational method CoSpRo on three benchmarks of spurious correlations, with controls that separate the content of the maps, the role of concept names and the role of location.

1 INTRODUCTION

An image can contain several predictive factors. In Waterbirds the bird category correlates with the background of the training images; in MetaShift 88% of the training cats sit indoors and 88% of the dogs outdoors. A representation that encodes the object together with its usual surroundings fails on birds in an uncommon environment and on cats on a lawn. Worst-group evaluation makes these failures visible when the average accuracy hides them (Sagawa et al., 2020). Self-supervised pretraining inherits the correlations of its unlabeled images: SimCLR rewards every feature that tells two images apart, the couch as much as the cat (Chen et al., 2020) and the augmentations leave open which changes of the context should preserve the identity of the object.

Our goal is a visual encoder whose downstream predictions stay useful when object–context associations change. Pretraining may use no target labels, no group annotations and no knowledge of which concept is spurious, since that knowledge differs between datasets: the background in Waterbirds, gender in CelebA with hair colour as the target. Deep feature reweighting shows that a linear head retrained on a small group-balanced set recovers worst-group accuracy whenever the representation encodes the core feature apart from the spurious one (Kirichenko et al., 2023). The question is therefore how to make a self-supervised encoder carry the cat as a feature of its own.

Our previous method, CoSpRo, approached this question through pairs of images. It removed the subspace of a concept group from frozen CLIP embeddings, kept the image pairs that became similar and transferred the resulting graph to a SimCLR student through a relational loss. CoSpRo improved Waterbirds and stayed at the SimCLR level on MetaShift and Spur-CIFAR10. It also relied on a prior: that the spurious attribute is a context concept whose removal reveals a useful relation.

SCD rests on a different observation. At the level of whole images, cat and couch arrive together on most training images; a student that sees only whole-image targets can satisfy them with one fused "cat-and-couch" feature. At the level of locations, the cat and the couch occupy different places in *every* image. A student that has to say "cat here, couch there" for every part of every view needs one code for the cat and another for the couch. Location separates what co-occurrence fuses and it does so for every concept alike, with no decision about which concept is spurious.

SCD turns the concepts of the training set into dense maps with a frozen CLIP model and asks every location of the student's last feature map to predict the concept values at its place in the view. A ridge regression fitted on one half of the batch scores the other half, so only concept directions shared across images lower the loss. During inference only the trained encoder and the downstream classifier remain.

Our contributions are:

- a prior-free spatial concept loss for self-supervised pretraining, built from named concepts that a frozen vision-language model locates on the training images;
- a cross-fitted regression readout that makes the loss reward concept codes shared across images and gives memorization of single images no reward;
- a controlled study on three datasets, with controls that separate the concept content of the maps (shuffled maps), named concepts against generic dense CLIP information (principal-component maps) and the role of location (image-level concept regression), a matched comparison with CoSpRo and a post-hoc audit of which concepts the student carries to unseen images.

2 Related Work

Dense and region-level self-supervised learning. SimCLR contrasts two views of an image against the other images in the batch (Chen et al., 2020) and pools the feature map before its projection head. DenseCL adds a contrastive loss between matched locations of the two views (Wang et al., 2021) and PixPro asks locations that cover the same image pixels to agree (Xie et al., 2021). DetCon pools the feature map inside unsupervised masks and contrasts the pooled regions across views (Hénaff et al., 2021). These methods supervise locations with the student's own features or with unnamed segments. SCD supervises every location with named concept values from a frozen teacher and keeps the SimCLR objective unchanged.

Dense concepts from vision-language models. CLIP supplies a shared image–text space (Radford et al., 2021). Its patch tokens carry little local information in the last block, where every patch attends to the same few global tokens (Zhou et al., 2022; Wang et al., 2024; Darcet et al., 2024). MaskCLIP aligns patch tokens with text once the last attention pooling is bypassed (Zhou et al., 2022); SCLIP replaces the query–key attention by a correlative self-attention (Wang et al., 2024); ClearCLIP removes the residual and feed-forward branches of the last block and uses query–query attention (Lan et al., 2024). All three need no training. SCD uses the ClearCLIP construction offline on the training images only and reads images through overlapping windows, as SCLIP and ClearCLIP do for open-vocabulary segmentation.

Distillation from vision-language teachers. Feature distillation from CLIP into a smaller student works with plain feature mimicry (Yang et al., 2024) and AM-RADIO distils several teachers with summary and spatial losses (Ranzinger et al., 2024). These students copy the teacher's full feature space. SCD distils a low-dimensional map of named concepts through a regression that is refitted in every batch, in the manner of differentiable closed-form solvers in meta-learning (Bertinetto et al., 2019; Lee et al., 2019). Its cross-fit follows the sample splitting of double machine learning (Chernozhukov et al., 2018), here applied to a training loss.

Concept representations. TCAV measures prediction sensitivity along directions learned from concept examples (Kim et al., 2018); ACE discovers visual concepts by clustering image segments in feature space (Ghorbani et al., 2019). Concept Bottleneck Models route predictions through annotated attributes (Koh et al., 2020); Label-Free Concept Bottleneck Models use language-derived concepts

without manual annotation (Oikarinen et al., 2023). SpLiCE expresses CLIP embeddings as sparse non-negative combinations of named text directions (Bhalla et al., 2024). SCD uses these coordinates to choose the vocabulary of concepts whose locations the student learns.

Spurious features and invariance in SSL. Contrastive learning can suppress harder features in favour of shortcut solutions; implicit feature modification counteracts this through perturbations of the contrastive comparison (Robinson et al., 2021). Content–style identifiability results explain how augmentation isolates invariant content under explicit assumptions about the data-generating process (von Kügelgen et al., 2021) and correlations between generative factors persist in the learned latents (Träuble et al., 2021). Learning-speed-aware sampling (LA-SSL) emphasizes slowly learned examples (Zhu et al., 2023); LateTVG prunes late layers of the encoder to form a second view that discards the easiest features (Hamidieh et al., 2024); background augmentations swap or remove salient backgrounds (Ryali et al., 2021). LA-SSL and LateTVG assume that the spurious feature is the easy one, background augmentation assumes that it is the background and CoSpRo assumes that it is a removable context concept. SCD makes no assumption about which concept is spurious.

Robustness with other supervision regimes. Group DRO uses annotated groups (Sagawa et al., 2020) and Just Train Twice upweights errors of an initial supervised classifier (Liu et al., 2021). GALS grounds a language description of the class in CLIP attention and guides a supervised network towards those pixels (Petryk et al., 2022), a descendant of explanation constraints from annotated masks (Ross et al., 2017). CoBalT discovers concepts without labels and uses them to balance supervised training (Arefin et al., 2024). SCD shares the use of language-grounded locations with GALS; it names no class and trains a separate encoder without labels. The supervised group-robust and zero-shot accuracies of these papers therefore offer no matched numerical comparison to our pipeline.

3 Problem Setting

Let $\mathcal{D} = \{x_i\}_{i=1}^{N}$ be unlabeled training images with stable sample identifiers. The learned representation is $f_\theta(x)$. A downstream classifier w predicts the target y from frozen features. For evaluation groups $g = (y, a)$ defined by the target and the spurious attribute a, we report empirical average accuracy and worst-group accuracy:

$$\text{Avg} = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}\{w(f_\theta(x_i)) = y_i\}, \qquad \text{WGA} = \min_g \frac{1}{n_g} \sum_{i:g_i=g} \mathbf{1}\{w(f_\theta(x_i)) = y_i\}. \tag{1}$$

Concept discovery, map construction and SSL use images and frozen representations. Labels and group annotations enter the downstream probe, the construction of its balanced subset and post-hoc analysis. "Label-free" refers to the concept and SSL stages; CLIP's external image–text pretraining and the supervised evaluation use labels. We add one restriction, the *prior-free* restriction: the method may use no knowledge of which concept is the target and which the spurious attribute. Every concept therefore enters the loss in the same way.

Notation. Throughout this report y denotes a class label and appears only in evaluation. The values that the student learns to predict during pretraining are *concept targets*, written t: soft scores of named concepts computed by the frozen teacher from the image pixels. They are neither class labels nor pseudo-labels of the class.

4 Method

The method in plain words. Before training starts, a large pretrained model that knows many words, CLIP, goes through every training photo once and colours it in like a colouring book: these squares are cat, these are couch, these are window. To make the squares small enough, each photo is first enlarged to 448×448 pixels and read through nine overlapping windows of the size CLIP was trained on, so every square of the final 14×14 grid covers one fourteenth of the photo's width and height. CLIP is then put away and never runs during training. Training is ordinary SimCLR: the student network sees two randomly cut, mirrored and recoloured pieces of each photo and learns to

recognize that both pieces come from the same photo. In addition, for every piece we know exactly which rectangle of the photo it shows, so we cut the same rectangle out of the photo's colouring page and shrink it to the student's own 7×7 grid. Every one of the 49 squares of the student now gets a question: "which concepts are here?". The student answers through a simple linear rule. The rule is learned on one half of the photos in the batch and checked on the other half, so a student that memorizes the answers of individual photos gains nothing; only a student that represents a cat in the same way in every photo can answer for photos the rule has not seen. Because the cat and the couch sit in different squares of every photo, the student has to learn what a cat looks like by itself and what a couch looks like by itself, although the two almost always appear together. The colouring pages are concept scores computed from the pixels, not class labels: nobody tells the student which concept is the class, and every concept is treated the same way. After training only the student is kept and a simple classifier on its features is evaluated on the hardest group of photos, such as cats outdoors.

The three stages. SCD has three stages (Figure 1). Stage 1 runs once per dataset before training: a frozen CLIP model names the concepts of the training images, groups synonyms into concept factors and computes a dense map of every factor for every image. Stage 2 is ordinary SimCLR training of a ResNet-18 from random weights with one additional loss: the student's last feature map predicts the part of the concept map that each view shows. Stage 3 evaluates the frozen student with a linear probe. CLIP runs only in Stage 1; training reads the stored maps.

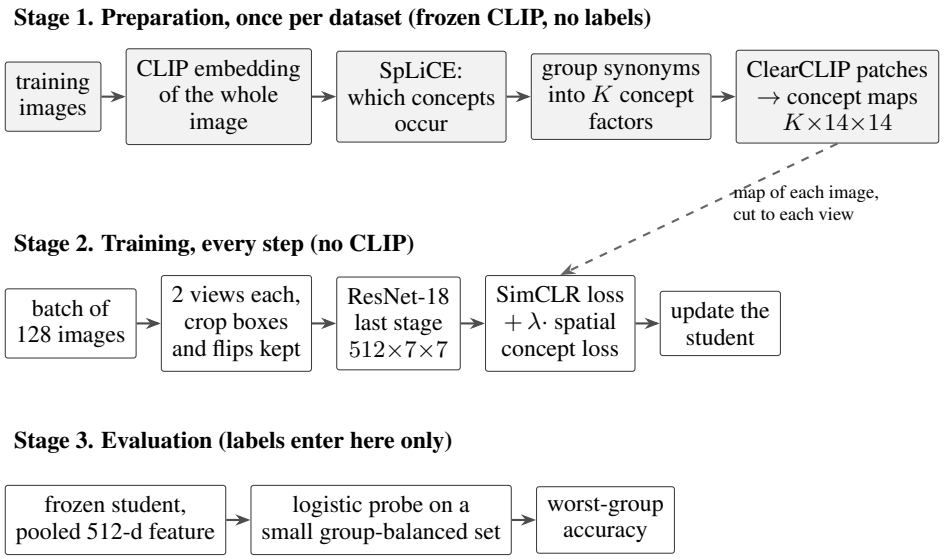


Figure 1: The three stages of SCD. After Stage 1 only the concept maps are kept; after Stage 2 only the ResNet-18 is kept. The maps, the projection head and every ridge fit are discarded. Concept discovery, map construction and SSL use no target or group annotations and no concept type.

4.1 Stage 1a: a concept vocabulary from SpLiCE

SpLiCE gives a vocabulary of the concepts that occur in the training set, without annotating which of them are spurious. Given an image x_i resized to 224 pixels, frozen CLIP produces the embedding $z_i \in \mathbb{R}^{512}$ of the *whole* image. With the fixed SpLiCE mean μ and a row-normalized text dictionary $D \in \mathbb{R}^{V \times 512}$ of V words, positive sparse regression gives

$$u_i = \mathrm{norm}(z_i - \mu), \qquad c_i \in \arg\min_{c \geq 0} \|u_i - D^\top c\|_2^2 + \lambda_1 \|c\|_1, \tag{2}$$

with $\mathrm{norm}(v) = v/\|v\|_2$ and $\lambda_1 = 0.25$. About nine coordinates of c_i are non-zero; a MetaShift image might give $c_i = 0.41\, e_{\text{cat}} + 0.30\, e_{\text{couch}} + 0.12\, e_{\text{blanket}} + \ldots$, where e_k is the coordinate of word k. SpLiCE runs on whole-image embeddings only. Its role is to decide *which* words describe this dataset and how often; the locations come later from patch embeddings.

4.2 STAGE 1B: CONCEPT FACTORS

Words of the dictionary often name one visual content several times (*cat*, *kitten*, *kitty*). If each word kept its own channel, synonyms would compete for the same pixels in the maps of Stage 1c and split the mass of one object over several names. We therefore group words into *concept factors*. The L1 penalty of SpLiCE activates one of two synonyms per image, so synonyms rarely co-activate and their centred dictionary directions can lie far apart. Raw CLIP text embeddings t_k^{text} keep synonyms close (cosine 0.85 to 0.94) and separate an object from its usual context (0.64 to 0.72). We keep words active on 0.2% to 95% of the training images and define the similarity of two words as

$$s_{k\ell} = \begin{cases} \cos(t_k^{\text{text}}, t_\ell^{\text{text}}) & \text{if } \rho_{k\ell} \geq 0.5, \\ -1 & \text{otherwise,} \end{cases} \tag{3}$$

where $\rho_{k\ell}$ is the correlation across training images of the responses $u_i^\top d_k$ and $u_i^\top d_\ell$. Average-linkage clustering on $1 - s_{k\ell}$, cut at similarity 0.85, forms the groups. Average linkage joins two clusters only when their members agree on average, which stops the chains that a union of pairwise matches forms through a bird and its background.

A concept factor F is one group, for example {kitty, kitten, cat}, {canine, dog} or {couch, sofa}. Its text direction is the normalized mean of its members' text embeddings,

$$d_F = \operatorname{norm}\left(\tfrac{1}{|F|}\textstyle\sum_{k \in F} \operatorname{norm}(t_k^{\text{text}})\right), \tag{4}$$

and we keep factors active on 2% to 90% of the training images. Waterbirds has $K = 242$ factors, MetaShift 130 and Spur-CIFAR10 147. The factors include the object classes and their contexts alike; nothing marks one of them as the class.

4.3 STAGE 1C: DENSE CONCEPT MAPS

Stage 1c computes, for every training image, where each factor sits. This needs three decisions: which patch embeddings to read, at which resolution and how to turn patch scores into a map.

Patch embeddings that match text. CLIP's vision transformer cuts the image into patches and keeps one token per patch. Training aligns only the class token with text. In the last block the query–key attention of every patch concentrates on the same few global tokens (Zhou et al., 2022; Wang et al., 2024; Darcet et al., 2024) and the residual branch adds this global signal to every patch, so a patch on the couch repeats the summary "cat, couch, room" of the whole image. ClearCLIP (Lan et al., 2024) changes only the last block. It drops the residual and feed-forward branches and replaces query–key attention by query–query attention, so a patch attends to patches that resemble it. With X the normalized tokens entering this block, the patch embeddings in the text space are

$$P = \operatorname{norm}\Big(\operatorname{LN}\big(\operatorname{softmax}(QQ^\top/\sqrt{d_h})\, V\, W_O\big)\, W_{\text{proj}}\Big), \qquad Q = XW_Q,\ V = XW_V, \tag{5}$$

applied per attention head and normalized per patch. Each patch u gets a unit vector $p(u) \in \mathbb{R}^{512}$ in the space of the factor directions d_F. ClearCLIP needs no training and no labels.

Resolution: why the image is read at 448 pixels. The teacher is OpenCLIP ViT-B/32 (Radford et al., 2021). Two properties of this model fix the resolution. First, its input is 224×224 pixels: it was trained at this size and its positional embeddings cover a 7×7 grid of patches. Second, a patch is 32 pixels wide, so a 224-pixel input gives $224/32 = 7$ patches per side. One CLIP patch then covers $\frac{1}{7}$ of the image side and CLIP cannot resolve anything smaller than its patch: the head of a cat and the cushion next to it share one vector.

The training images of MetaShift and Waterbirds have arbitrary sizes, so every image has to be resized to some fixed size in any case. Table 1 lists the options. Feeding the image at 224 pixels gives a 7×7 map. Feeding 448 pixels in one pass gives 14×14 patches, but CLIP then runs at a size it was not trained on and its positional embeddings must be interpolated. SCD resizes every image to 448×448 pixels and reads it through nine windows of 224×224 pixels at offsets $\{0, 112, 224\}$ in each direction (Figure 2). Every window is a native 224-pixel CLIP input and yields 7×7 patches of 32 pixels, while 32 pixels are now $\frac{1}{14}$ of the image side. The map resolution doubles without any input that CLIP has not seen in training. The same sliding-window reading is the evaluation protocol of SCLIP and ClearCLIP for open-vocabulary segmentation (Wang et al., 2024; Lan et al., 2024).

Table 1: How the size of the image fed to CLIP ViT-B/32 determines the concept map. The patch is always 32 pixels.

Input to CLIP	Patches	One patch covers	Consequence
original image	–	–	arbitrary sizes; CLIP needs a fixed input
224×224, one pass	7×7	$\frac{1}{7}$ of the side	a small crop covers about 3×3 patches
448×448, one pass	14×14	$\frac{1}{14}$ of the side	positional embeddings interpolated beyond training
448×448, nine 224-px windows	$9 \times (7 \times 7)$	$\frac{1}{14}$ of the side	native input in every window; used by SCD

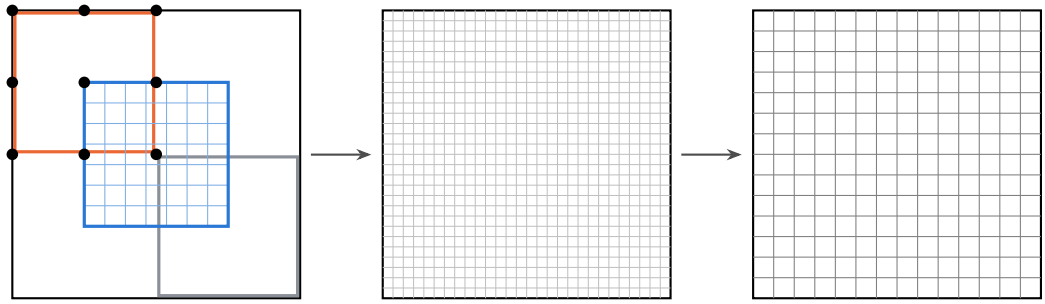


Figure 2: From windows to the 14×14 map. Left: the nine windows have their top-left corners at pixels 0, 112 and 224 in each direction (dots); three of them are drawn, the top-left (orange), the centre (blue, with its 7×7 patches) and the bottom-right (grey). Neighbouring windows overlap by half, so their patches fall on a 16-pixel grid (middle) and a cell is covered by one, two or four windows. Right: the averaged scores pooled to the 14×14 map.

Where 14×14 comes from. For every window and every patch u we compute the cosine with every factor direction, $s_F(u) = p(u)^\top d_F$. Windows overlap by half their width, so the patches of neighbouring windows are shifted by 112 pixels, which is three and a half patches. Their borders therefore fall on a grid of 16-pixel cells, $448/16 = 28$ cells per side. Each patch spreads its scores onto its 2×2 cells, scores are averaged where windows overlap and the 28×28 grid is average-pooled to 14×14, one cell per patch width. The averaging over one, two or four windows also reduces the noise of single patches.

Why the map stops at 14×14. The resolution of the map is set by the student, which sees views of 20% to 100% of the image area (Section 4.4) and has a 7×7 feature map. A view that covers the whole image spans 14×14 cells, so each of its 7×7 targets averages 2×2 cells. The smallest view spans $\sqrt{0.2} \approx 0.45$ of the image side, which is $0.45 \times 14 \approx 6$ cells per side: close to one map cell per student location, so even the smallest crop receives real, separately computed content. A 7×7 map would give the smallest crop about 3×3 cells, stretched over 7×7 locations and neighbouring student locations would receive identical blurred targets exactly on the views where an object fills half the view. A 28×28 map adds no resolution of its own, since its 16-pixel cells come from averaging 32-pixel patches and it takes four times the memory: the 14×14 maps take 87 MB for MetaShift and 455 MB for Waterbirds, against about 1.8 GB for Waterbirds at 28×28. Training holds the maps in memory.

The square resize keeps maps and views aligned. Resizing to 448×448 changes the aspect ratio of the image. The map covers the whole image and the crop box of a view is stored as fractions of the image side (Section 4.4). A fraction such as "from 30% to 70% of the width" is the same before and after any resize, so the box lands on the right cells of the map whatever the original size and shape of the image.

From scores to a map. At every cell u of image i the map distributes one unit of mass over the factors,

$$M_{iF}(u) = \frac{\exp\big(s_{iF}(u)/\tau_M\big)}{\sum_{G=1}^{K} \exp\big(s_{iG}(u)/\tau_M\big)}, \qquad \tau_M = 0.01. \tag{6}$$

The temperature is CLIP's own scale: CLIP multiplies its cosines by a learned logit scale of 100 before its zero-shot softmax (Radford et al., 2021). Cosines of one patch with different words differ by a few hundredths and this scale lets one word win a cell. For illustration, a cell with cosines 0.27 with *cat*, 0.25 with *blanket* and 0.21 with *couch* becomes about 88% cat, 12% blanket and 0.2% couch; at $\tau_M = 1$ the three would receive 34%, 33% and 32%. When training loads the maps, every factor is standardized over all images and cells, $\tilde{M}_{iF} = (M_{iF} - m_F)/s_F$, so that frequent factors such as blanket or lawn do not dominate the squared error of Section 4.6. The maps are stored once in float16 and take 40 seconds to compute for MetaShift on one GPU. Figure 3 shows maps of four MetaShift images.

4.4 Stage 2a: views, crop boxes and the student feature map

Training is ordinary SimCLR on a ResNet-18 from random weights. Every image of a batch of 128 gives two views. The transform of a view applies a random crop of 20% to 100% of the image area with aspect ratio between 3:4 and 4:3, resized to 224×224; a horizontal flip with probability 0.5; colour jitter with probability 0.8 and greyscale with probability 0.2. Crop and flip are the only operations that move pixels, so the transform records for every view its crop box as fractions of the image (top a, left b, height η, width ω) and its flip flag. These records tell which part of the concept map each view shows.

The ResNet-18 (He et al., 2016) halves the resolution five times and maps a view to $H \in \mathbb{R}^{512 \times 7 \times 7}$ at the output of its last residual stage, since $224/2^5 = 7$. Global average pooling gives the 512-dimensional feature h that the probe reads after training and an MLP head maps h to the 128-dimensional input of the SimCLR loss. The concept loss reads H before pooling. Location (r, c) of H lies over the 32-pixel square (r, c) of the view, so each of the 49 vectors can be asked about its own part of the view; class activation maps rest on the same property of the last feature map (Zhou et al., 2016). After pooling, a cat and a couch in the same view become one vector; before pooling they still

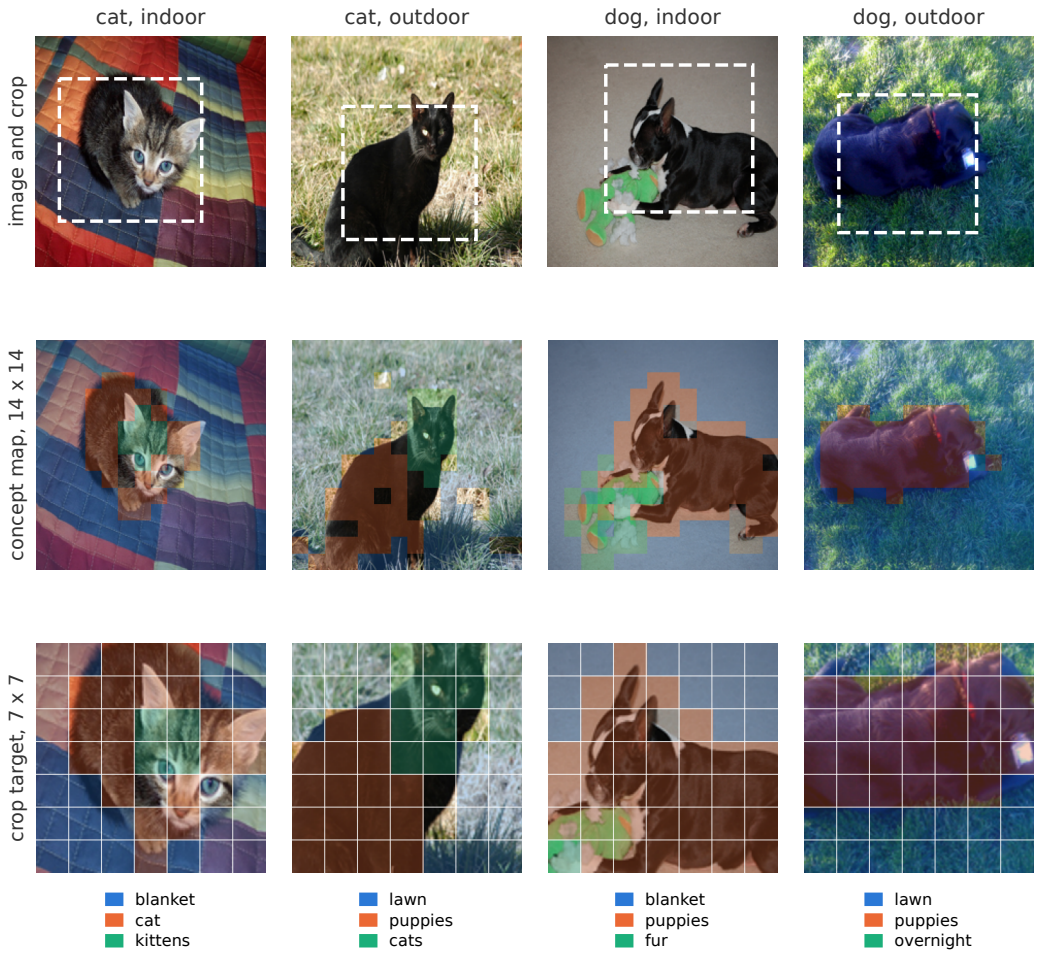


Figure 3: Concept maps of MetaShift training images, one per group. Top: the image and one SimCLR crop (dashed). Middle: the factor that wins each cell of the 14×14 map, coloured for the image's three most frequent factors. Bottom: the crop's targets after warping to the student's 7×7 feature map (Section 4.5). The teacher makes mistakes: part of the black cat in the second image is labelled "puppies". Class and group labels play no part in map construction or training.

occupy different locations. This is why the concept loss reads the feature map and leaves the pooled feature to SimCLR. The probe reads the pooled feature of the same backbone that the loss shapes.

The student's 7×7 grid and the 7×7 patch grid of a CLIP window have the same size for different reasons ($224/2^5$ and $224/32$). The two grids never meet: CLIP's grids are assembled into the 14×14 map of the whole image and the warping below resamples that map to the student's grid of each view.

4.5 Stage 2b: warping the map to each view

For a view with box (a, b, η, ω), the box in map coordinates is $[14b,\ 14(b+\omega)] \times [14a,\ 14(a+\eta)]$. RoIAlign (He et al., 2017) divides it into 7×7 bins and averages 2×2 bilinear samples of $\tilde{M}_i$ inside each bin; a flipped view gets its targets mirrored along the width. The result is the target tensor of view v,

$$T^{(v)} = \text{flip}^{\,\phi_v}\Big(\text{RoIAlign}\left(\tilde{M}_i,\ \text{box}_v,\ 7 \times 7\right)\Big) \in \mathbb{R}^{K \times 7 \times 7}, \tag{7}$$

with $\phi_v \in \{0, 1\}$ the flip flag. Cell (r, c) of $T^{(v)}$ describes exactly the part of the view that student location (r, c) looks at (Figure 4). RoIAlign reads the map at fractional positions, so a crop that starts inside a cell receives the correct values.

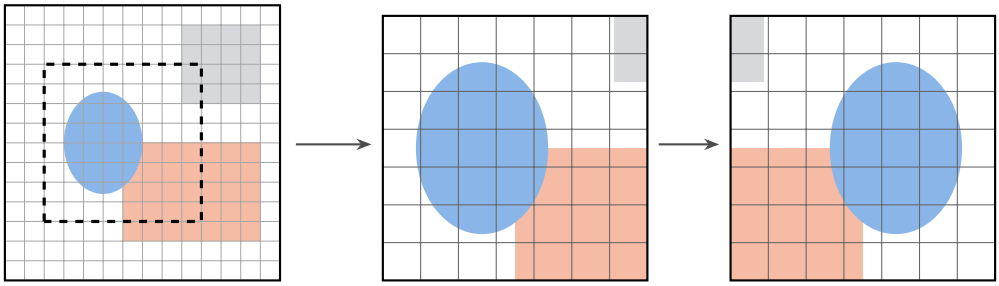


Figure 4: Warping the map to one view (toy scene). All K channels of the map are cut and resampled in the same way; the figure colours each cell by its winning factor: cat, couch and window.

4.6 Stage 2c: the cross-fitted spatial concept loss

Question–answer pairs. Every location u of every view v forms one pair: a unit-norm feature row $r = \text{norm}(H^{(v)}(u)) \in \mathbb{R}^{512}$, what the student sees there and a target $t = T^{(v)}(u) \in \mathbb{R}^K$, which concepts the teacher's map places there. The 128 images of a batch are split into halves A and B of 64 images, with both views of an image in the same half. Each half holds $64 \times 2 \times 49 = 6{,}272$ pairs. Stacking them gives $R_A \in \mathbb{R}^{6272 \times 512}$ and $T_A \in \mathbb{R}^{6272 \times K}$; Table 2 shows a few rows for MetaShift.

Table 2: Rows of half A (MetaShift, $K = 130$). Feature values and targets are illustrative; targets are standardized, so 0 is the average presence of a factor and positive values mark cells where it is more present than average.

Source of the row	$r \in \mathbb{R}^{512}$ (student)	$t \in \mathbb{R}^{130}$ (map): cat, couch, lawn, …
cat on couch, view 1, location (3,2)	$(0.04, -0.11, \ldots)$	$(\ 2.1, -0.3, -0.4, \ldots)$
cat on couch, view 1, location (5,5)	$(0.02,\ \ 0.07, \ldots)$	$(-0.5,\ \ 2.4, -0.4, \ldots)$
cat on couch, view 2, location (3,4)	$(0.05, -0.09, \ldots)$	$(\ 1.8, -0.1, -0.4, \ldots)$
dog on lawn, view 1, location (6,1)	$(-0.03,\ \ 0.01, \ldots)$	$(-0.5, -0.3,\ \ 2.7, \ldots)$
	… 6,272 rows in total	

Why a readout is needed. The 512 coordinates of the student do not correspond to the K factors and the student should be free to arrange its feature space. A linear map $W \in \mathbb{R}^{512 \times K}$ translates features into concept values: the predicted value of *cat* at a location is the projection of its feature

row on the cat column of W. A linear readout matches the evaluation, where a linear probe reads the features, so concepts have to become *linearly* readable.

Why the readout is refitted and cross-fitted. A readout trained by gradient descent alongside the student could store a target per image: a deep network can fit arbitrary targets on a small dataset (Zhang et al., 2017) and MetaShift has 1,700 training images. A student that recognizes "this is image 371" and stores its answers lowers such a loss without learning any concept. SCD therefore fits the readout in closed form on half A and scores it on half B, whose images the fit has never seen. With Ω_A the diagonal matrix of per-row loss weights, $\bar{r}_A$ and $\bar{t}_A$ the weighted means of rows and targets in A and $\bar{R}_A = R_A - \mathbf{1}\bar{r}_A^\top$, the weighted ridge regression is

$$W_A = \left(\bar{R}_A^\top \Omega_A \bar{R}_A + \rho I\right)^{-1} \bar{R}_A^\top \Omega_A \left(T_A - \mathbf{1}\bar{t}_A^\top\right) \in \mathbb{R}^{512\times K}, \qquad \rho = 1, \tag{8}$$

one 512×512 linear system (Hoerl and Kennard, 1970). The fit predicts every location of B,

$$\hat{T}_B = \left(R_B - \mathbf{1}\bar{r}_A^\top\right) W_A + \mathbf{1}\bar{t}_A^\top, \tag{9}$$

and the reverse fit W_B predicts A (Figure 5). The loss is the held-out squared error, averaged over the two directions,

$$\mathcal{L}_{\text{spatial}} = \frac{1}{2} \sum_{(S,S')\in\{(A,B),(B,A)\}} \frac{\sum_{v\in S'} \omega_v \frac{1}{49K} \left\|\hat{T}^{(v)} - T^{(v)}\right\|_2^2}{\sum_{v\in S'} \omega_v}, \tag{10}$$

where $\hat{T}^{(v)}$ collects the predictions of the 49 locations of view v by the fit on the other half and ω_v is its loss weight. Gradients flow through the prediction and through the solve for W_A into H and the ResNet weights (Bertinetto et al., 2019). The readout has no parameters of its own; it is solved again in every step from the current features and discarded afterwards.

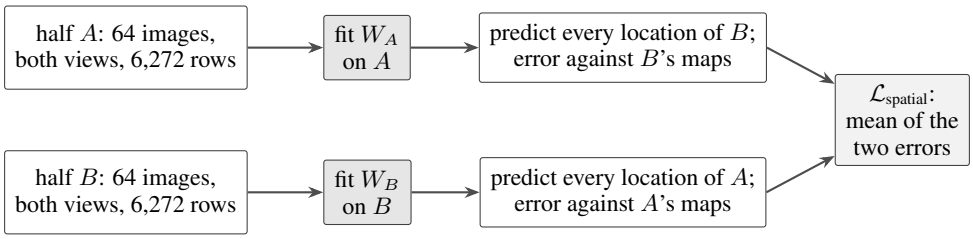


Figure 5: Cross-fitting. Each readout is scored only on images it was not fitted on. Gradients pass through the solve and the prediction into the backbone.

Two ways to lower the loss. Consider two students. The first memorizes: it gives every image i its own feature direction e_i, orthogonal to the directions of other images and W_A maps each e_i of half A to that image's targets. On a location of an image j in half B, e_j is orthogonal to every direction that W_A was fitted on, so the prediction collapses to about $\bar{t}_A$, the average target of A. Since the targets are standardized, this prediction has an error close to 1, the error of predicting the mean. Memorization earns nothing. The second student uses shared codes: a cat direction that appears on cat locations of every image and a couch direction on couch locations. W_A learns "cat direction $\mapsto$ cat value" from the cats of A and the same rule holds for the cats of B. Only this second kind of feature lowers $\mathcal{L}_{\text{spatial}}$. Because the targets have unit variance, $1 - \mathcal{L}_{\text{spatial}}$ is the held-out explained variance, which we log during training.

Design choices of the loss.

- **Unit-norm rows with an absolute ridge.** A student that could shrink its features would weaken the ridge penalty and let the readout overfit. Unit rows fix the scale and $\rho = 1$ keeps the 512×512 system well conditioned.
- **Both views of an image in one half.** Otherwise half B would contain the second view of an image of half A and recognizing the image would pay off again.
- **Locations as rows.** Each half has 6,272 rows for 512 unknowns per factor, a well-posed fit at batch size 128. A readout on pooled features would have 128 rows per half.
- **Precision.** The network runs under float16 autocast; the solve and the loss run in float32.

Hold-out images. A fixed fifth of the training images (340 on MetaShift) carries loss weight $\omega_v = 0$: their rows shape no readout and enter no error, while SimCLR trains on them as usual. After training they measure whether the student carries concepts to images outside the concept loss (Section 7.5).

4.7 THE COMPLETE OBJECTIVE

The SimCLR term is unchanged. With g_v the normalized projection of view v, v^+ the other view of the same image and $\tau = 0.05$, the NT-Xent loss (van den Oord et al., 2018; Chen et al., 2020) over the $2B = 256$ views of a batch is

$$\mathcal{L}_{\mathrm{SimCLR}} = -\frac{1}{2B}\sum_{v=1}^{2B} \log \frac{\exp(g_v^\top g_{v^+}/\tau)}{\sum_{w \neq v} \exp(g_v^\top g_w/\tau)}. \tag{11}$$

Each view has one positive (the other view of its image) and 254 negatives, among them views of other cats. SimCLR rewards every feature that separates images and has no reason to separate a cat from the couch it usually sits on; the spatial loss adds that reason. The training objective is

$$\mathcal{L} = \mathcal{L}_{\mathrm{SimCLR}} + \lambda(e)\,\mathcal{L}_{\mathrm{spatial}}, \qquad \lambda(e) = \begin{cases} 0, & e \leq 10, \\ 5\,(e-10)/10, & 10 < e < 20, \\ 5, & e \geq 20, \end{cases} \tag{12}$$

with e the epoch. SimCLR first forms usable features, then the concept loss rises to its full weight.

5 SCD AND COSPRO

Both methods use the same frozen CLIP/SpLiCE vocabulary, the same SimCLR student and the same evaluation. They differ in what the teacher computes and in how it reaches the student (Table 3).

The teacher signal. CoSpRo computes *relations between images*. For a concept group G it removes the span Q_G from the centred CLIP embeddings, $u_i^{-G} = \mathrm{norm}(u_i - Q_G Q_G^\top u_i)$, keeps pairs whose similarity rises, filters them by activation contrast, residual concept agreement and synthetic nulls and stores a sparse graph with a distribution $p_T(j \mid i)$ over donor images (Figure 6). SCD computes a *description of each location of each image*: a concept map. CoSpRo's signal is one number per image pair; SCD's signal is K numbers per cell, about $196K$ values per image.

The prior. CoSpRo searches for relations that appear when a concept is removed. This helps when the spurious attribute is a removable context concept and the class survives the removal. The same projection can remove object information: on Waterbirds 46% of the graph mass of waterbirds on land connects images of the other class. SCD removes nothing and ranks no concept above another. The cat and the couch enter the loss as two factors with the same weight.

How the student is shaped. CoSpRo acts on the pooled features: graph-aware batches place teacher neighbours together and a relational KL loss asks the student's similarity distribution to reproduce p_T. Only anchors with a donor in the batch contribute. SCD acts on the feature map before pooling: every location of every view of every image contributes in every step.

Memorization. CoSpRo's targets are relations between specific images, which the student could satisfy by memorizing those pairs. SCD's cross-fit scores every readout on images it has not seen, so only codes shared across images lower the loss; the hold-out images confirm that the concepts transfer (Section 7.5).

Cost and scope. CoSpRo's graph construction evaluates candidate pairs for every concept group and calibrates every group against null groups. SCD's maps take 40 seconds for MetaShift and add no tuned threshold beyond the factor grouping. CoSpRo improved Waterbirds; SCD improves Waterbirds, MetaShift and Spur-CIFAR10 (Section 7.2).

Table 3: CoSpRo and SCD side by side.

	CoSpRo	SCD
Teacher output	sparse graph of image pairs with weights	dense map of K concept factors per image
Teacher computation	subspace removal, filters, null calibration	ClearCLIP patch scores, 9 windows, softmax
Assumption about the spurious attribute	a removable concept; its removal reveals same-class pairs	none; all concepts enter alike
Student signal	relational KL on pooled features	cross-fitted regression on the 7×7 feature map
Batches	graph-aware sampler	ordinary random batches
Rows per step	anchors with a donor in the batch	$2 \times 6{,}272$ locations
Protection against memorization	none	readout scored on the other half of the batch
Effect on WGA	Waterbirds only	Waterbirds, MetaShift, Spur-CIFAR10

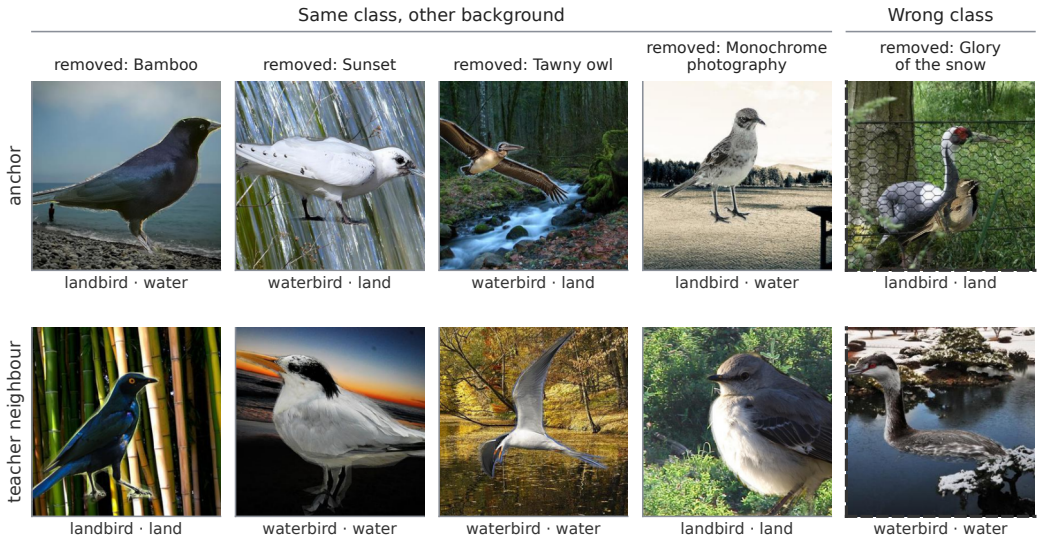


Figure 6: CoSpRo's teacher signal for comparison with Figure 3: retained graph edges on Waterbirds. Each column pairs an anchor (top) with its teacher neighbour (bottom) and names the concept whose removal produced the edge. The dashed pair on the right connects images of different classes. SCD replaces these pairwise relations by per-location concept maps.

6 EXPERIMENTAL PROTOCOL

We evaluate on Waterbirds (4,795 training images; Sagawa et al., 2020), MetaShift cats and dogs (1,700 training images; Liang and Zou, 2022) and Spur-CIFAR10, a CIFAR-10 variant (Krizhevsky, 2009) in which a 2-pixel coloured line agrees with the class on 95% of the 45,000 training images and on 10% of the 5,000 validation images. Each dataset gets its own concept factors and maps from its training images. On Spur-CIFAR10 the 32-pixel images are upsampled to 448 pixels for the maps and the student's last feature map is 4×4.

We pretrain a randomly initialized ResNet-18 for 500 epochs with batch size 128, SGD with learning rate 0.01 decayed by 0.1 at epochs 350, 400 and 450, weight decay 10^{-4} and SimCLR temperature 0.05, with the same augmentations for every method. The protocol follows CoSpRo, LateTVG and LA-SSL, which pretrain on the dataset's own images, so that every method sees the same images and budget. The concept loss uses ridge $\rho = 1$, maximum weight 5 and the 20% hold-out. We report six student seeds on Waterbirds and MetaShift and two on Spur-CIFAR10.

Frozen student features are evaluated every 25 epochs with the same logistic probe (L2 coefficient 0.001), fitted on a group-balanced subset of the training split: 224 images on Waterbirds, 400 on MetaShift and 1,400 on Spur-CIFAR10. Group labels are used for this probe and its evaluation only. Unless stated otherwise each value is the mean of the last four probes (epochs 425 to 500), which damps single-probe noise on the small MetaShift validation groups (72 images each, so one image is 1.4 points of a group's accuracy). For the matched comparison with CoSpRo we also report the single probe at epoch 500, the protocol of the CoSpRo paper. We report mean and sample standard deviation across student seeds. All SCD results are validation results; held-out test runs with a locked probe are in progress.

7 RESULTS

7.1 SCD AGAINST SIMCLR ON THREE DATASETS

Table 4: Validation evaluation, mean of the last four probes (SSL epochs 425–500). Waterbirds and MetaShift use seeds 1–6, Spur-CIFAR10 seeds 1–2. Shuffled maps give every image another image's maps. Values are percentages (mean ± sample standard deviation).

	Waterbirds		MetaShift		Spur-CIFAR10	
Method	Avg.	WGA	Avg.	WGA	Avg.	WGA
SimCLR	52.44 ± 1.07	46.66 ± 1.89	55.16 ± 3.99	44.33 ± 7.31	59.55 ± 0.23	15.63 ± 1.04
Shuffled maps	52.79 ± 2.71	47.50 ± 2.49	56.08 ± 2.23	45.08 ± 2.91	55.90 ± 0.71	17.25 ± 1.98
SCD	$\mathbf{60.32 \pm 2.66}$	$\mathbf{52.76 \pm 4.68}$	$\mathbf{61.15 \pm 2.95}$	$\mathbf{51.10 \pm 5.44}$	$\mathbf{69.32 \pm 1.01}$	$\mathbf{38.13 \pm 0.51}$

SCD achieves higher mean WGA than SimCLR by 6.10 points on Waterbirds, 6.77 on MetaShift and 22.50 on Spur-CIFAR10 (Table 4). The corresponding average-accuracy differences are $+7.88$, $+5.99$ and $+9.77$ points. SCD is ahead of SimCLR on five of six paired seeds on Waterbirds and MetaShift and on both Spur-CIFAR10 seeds (Figure 7, which also shows CoSpRo). The shuffled-map control trains the same loss on maps of other images: it keeps the statistics of the targets and removes their link to the image content, in the manner of the control tasks of the probing literature (Hewitt and Liang, 2019). It stays within one point of SimCLR's WGA on Waterbirds and MetaShift and within 1.6 points on Spur-CIFAR10, so the gain comes from the content of each image's own maps. Seed variation is large on MetaShift, whose validation groups hold 72 images each. The weight $\lambda_{\max} = 5$ and the hold-out were chosen on the validation splits of MetaShift and Waterbirds with seeds 1–4; seeds 5 and 6 ran afterwards with the settings fixed.

7.2 SCD AGAINST COSPRO

The SimCLR runs of seeds 1–4 on Waterbirds are shared by the CoSpRo study and this study, so the CoSpRo paper's validation protocol (single probe at epoch 500, seeds 1–4) gives a matched comparison (Table 5). SCD reaches 53.06 WGA against 50.00 for CoSpRo and 45.54 for SimCLR; the average accuracy rises from 53.46 to 60.28. The paired WGA differences between SCD and

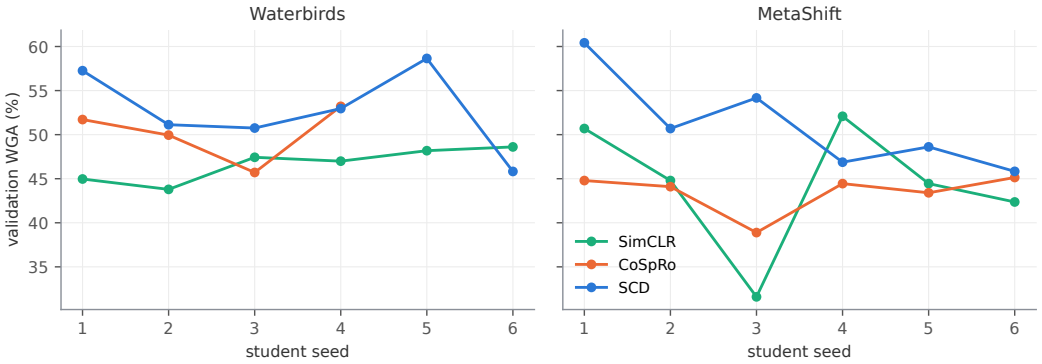


Figure 7: Validation WGA per student seed for SimCLR, CoSpRo and SCD, mean of the last four probes. CoSpRo has seeds 1–4 on Waterbirds; on MetaShift seeds 1–6 of its ten are shown.

CoSpRo are $+6.28$, $+1.13$, $+5.05$ and -0.21 points. The comparison favours SCD on average and on three of four seeds, with a margin of about one seed standard deviation.

Table 5: Matched Waterbirds comparison: validation, single probe at SSL epoch 500, seeds 1–4, the protocol of the CoSpRo paper. Per-seed values are WGA; the SimCLR runs are shared by both studies.

Method	Avg.	WGA	Seed 1	Seed 2	Seed 3	Seed 4
SimCLR	52.11	45.54 ± 1.71	44.54	43.68	47.32	46.62
CoSpRo + KL	53.46	50.00 ± 3.23	51.61	50.00	45.49	52.89
SCD	**60.28**	$\mathbf{53.06 \pm 3.35}$	57.89	51.13	50.54	52.68

Across datasets the difference is larger (Table 6). CoSpRo stayed at the SimCLR level on MetaShift over ten seeds (42.9 against 44.4) and reached 19.6 WGA on one Spur-CIFAR10 seed, while SCD gains 6.8 and 22.5 points there. On the held-out Waterbirds test split CoSpRo reached 50.64 ± 1.97 against 46.82 ± 3.65 for SimCLR; the matched test runs of SCD are in progress, so the test comparison remains open.

Table 6: Validation WGA of SimCLR, CoSpRo and SCD on the three datasets, mean of the last four probes. Seeds in brackets. The MetaShift CoSpRo study ran its own ten SimCLR seeds, which reach 44.41 WGA.

Method	Waterbirds	MetaShift	Spur-CIFAR10
SimCLR	46.66 ± 1.89 (6)	44.33 ± 7.31 (6)	15.63 ± 1.04 (2)
CoSpRo + KL	50.14 ± 3.25 (4)	42.92 ± 2.62 (10)	19.6 (1)
SCD	$\mathbf{52.76 \pm 4.68}$ (6)	$\mathbf{51.10 \pm 5.44}$ (6)	$\mathbf{38.13 \pm 0.51}$ (2)

The differences of Section 5 explain this pattern. CoSpRo's projection reveals same-class pairs when the removed concept is the background and the bird survives the removal, which holds best for Waterbirds. SCD needs no such condition: its targets describe every concept of every location and its cross-fit keeps the signal from turning into memorized pairs on the 1,700 MetaShift images.

7.3 Do named concepts and their locations add value beyond CLIP?

Two controls isolate the parts of SCD. The principal-component control replaces the concept maps by as many principal components of the dense ClearCLIP patch embeddings as there are factors, computed over the same cells and trained with the same loss. It keeps the dense CLIP information of the teacher and removes the concept names. The image-level control regresses each image's SpLiCE factor activations from the pooled features with the same cross-fitted ridge, weight and hold-out. It keeps the concepts and removes their locations.

Table 7: Controls, validation at epochs 425–500, seeds 1–4. Mean ± sample standard deviation in percent. All share the backbone, optimization, hold-out and probing protocol. †Seeds 1–2. ‡Factors from a 5% frequency band (70 on Waterbirds, 24 on MetaShift).

	Waterbirds		MetaShift	
Method	Avg.	WGA	Avg.	WGA
SimCLR	52.25 ± 1.21	45.80 ± 1.71	54.80 ± 4.81	44.79 ± 9.35
Shuffled maps	52.60 ± 3.48	46.73 ± 2.70	55.62 ± 2.27	45.31 ± 3.38
CLIP principal-component maps	58.07 ± 0.94	53.57 ± 1.93	56.29 ± 2.23	44.44 ± 7.37
Image-level concept regression‡	57.96 ± 3.95	50.37 ± 8.98	61.68 ± 0.18†	47.22 ± 3.93†
SCD	60.15 ± 2.46	53.02 ± 2.98	62.22 ± 2.97	53.04 ± 5.75

On MetaShift, SCD has 8.59-point higher mean WGA than the principal-component maps and is ahead on all four seeds; the paired differences are $+15.97$, $+1.39$, $+4.17$ and $+12.85$ points (Table 7). The principal-component maps stay at the SimCLR level there. On Waterbirds the two tie (-0.54 WGA, $+2.07$ Avg.): birds against land and water are among the strongest directions of CLIP's patch embeddings, so its leading components carry them, while cats against dogs and couches against lawns are weaker directions that only the named concepts single out. SCD exceeds the image-level regression by 2.66 WGA points on Waterbirds and 8.33 on the two MetaShift seeds; on Spur-CIFAR10 the image-level regression reaches 29.39 WGA against 38.13 (two seeds each). Image-level heads also store the targets of MetaShift's training images: at weight 10 the concept score on images inside the loss reaches 0.60 while the score on held-out images falls to 0.04. The spatial loss shows no such gap (Section 7.5).

A concept ablation tests which concepts carry the MetaShift gain. It uses word lists we chose and lies outside the method. Over seeds 1–4, leaving out the 21 animal factors (cat, dog, breeds, pet, paws, fur) gives 46.61 ± 5.75 WGA and leaving out the 20 scene factors (couch, lawn, bed, window, desk, bench and others) 47.92 ± 2.81, against 53.04 ± 5.75 with all factors and 45.31 ± 3.38 with shuffled maps. Both ablations lose most of the gain and differ by less than the seed noise: the gain needs the object and the scene concepts together, which is what lets the student tell them apart.

7.4 Training dynamics and groups

The worst-group advantage of SCD holds across training (Figure 8). On MetaShift its smoothed mean WGA leads SimCLR and CoSpRo at every probe, while CoSpRo falls below SimCLR from epoch 175 onward. On Waterbirds CoSpRo rises first and leads from epoch 50 to 125; the two alternate up to epoch 200 and from epoch 225 SCD stays ahead to the end of training. SCD leads both methods in average accuracy at every probe on both datasets, while CoSpRo stays within two points of SimCLR.

The gain extends to the minority groups (Figure 9): on MetaShift cats outdoors rise from 47.0 to 51.2 and dogs indoors from 52.8 to 57.2; on Waterbirds landbirds on water rise from 53.6 to 65.1. SCD raises every group on both datasets. CoSpRo raises the land groups of Waterbirds (landbirds on land 47.6 to 51.0, waterbirds on land 53.8 to 56.6) and leaves landbirds on water at 52.3; on MetaShift its groups stay within the seed spread of SimCLR.

7.5 What does the student learn from the maps?

The maps place the animal factors on the animal and the scene factors on its surroundings, whatever the group of the image (Figure 3). On MetaShift blankets, lawn, cats and fields win the most cells (7.4%, 6.6%, 5.8% and 5.6%); on Waterbirds bamboo, forest and tree stand win 37% of the cells and the factor *water bird* 6.2%.

Concepts transfer to unseen images. After training, a ridge regression from the pooled student features to each factor's image-level response explains, on the held-out fifth of the training images, 0.22 of the variance on MetaShift and 0.26 on Waterbirds, against 0.13 and 0.17 with shuffled maps (Table 8). The scores on images inside the concept loss agree (0.23 and 0.26), so the spatial loss learns shared concepts and stores no per-image answers. During training the held-out explained variance $1 - \mathcal{L}_{\text{spatial}}$ starts near 0.4, since maps are smooth and partly predictable from position and

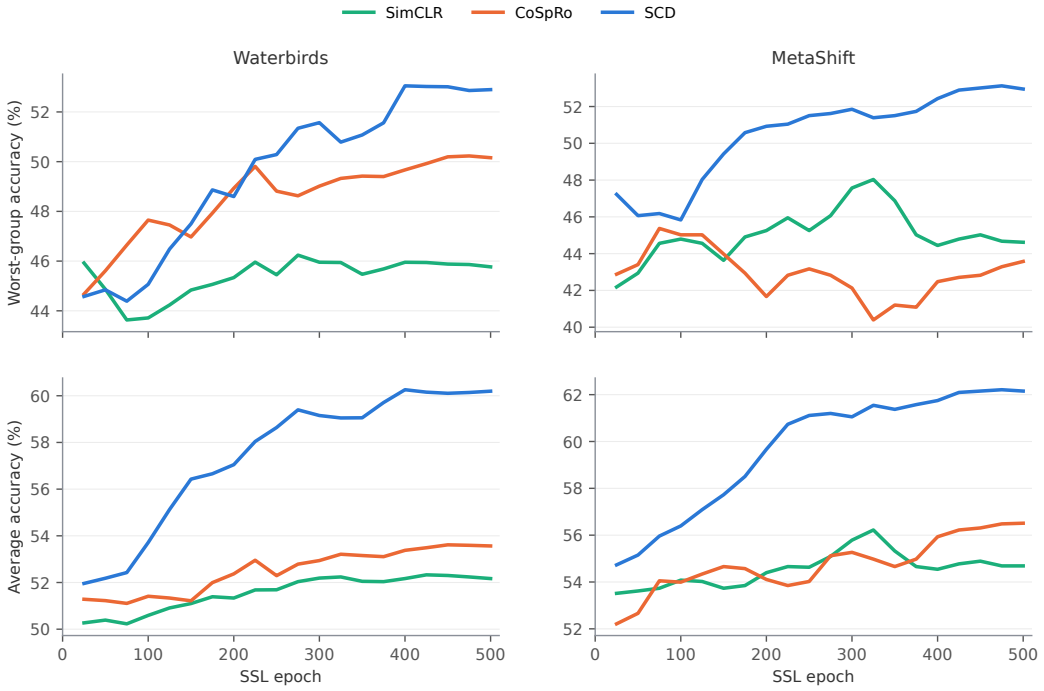


Figure 8: Validation accuracy during SSL training for SimCLR, CoSpRo and SCD. Each line is the mean over seeds 1–4 of the periodic linear probes, run every 25 epochs, smoothed with a three-point moving average. Top: worst-group accuracy. Bottom: average accuracy.

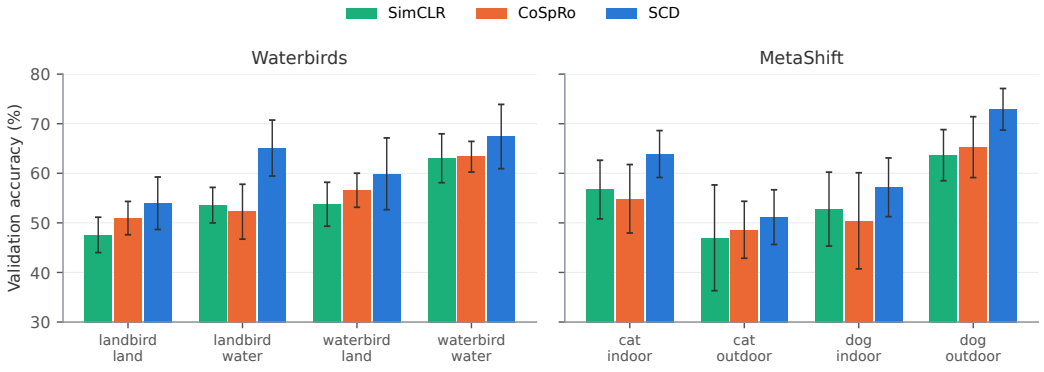


Figure 9: Validation accuracy per (class, attribute) group at the epoch-500 probe, mean and sample standard deviation over seeds (SimCLR and SCD: 6; CoSpRo: 4 on Waterbirds, 10 on MetaShift). Labels are used only for evaluation.

low-level statistics and ends at 0.44 to 0.48 for real maps against 0.39 to 0.41 for shuffled maps; only its rise above the shuffled control reflects concepts.

Concepts stay tied to their context. Post-hoc labels show that the student strengthens the core feature without separating it from its usual context. The student's *cat* factor scores 0.47 on unseen cats indoors and 0.07 on unseen cats outdoors, where CLIP features score 0.92 and 0.88. The *water bird* factor drops in the same way on waterbirds on land. Each factor's residual on the other factors, the part a fused direction cannot predict, scores 0.00 in the student against 0.15 to 0.17 for CLIP. A probe for the spurious attribute reaches 90.50 ± 0.79 WGA on Waterbirds for SCD against 89.50 ± 1.05 for SimCLR, so the background stays as readable as before. A location of the 7×7 feature map is computed from a large part of the view (Luo et al., 2016), so the location over the cat also sees the couch; the cheapest predictor of "cat here" combines local evidence with the scene, which agrees with the cat on 88% of the MetaShift training images. SCD therefore raises WGA through a stronger core feature that the balanced probe can use (Kirichenko et al., 2023), while the spurious attribute stays readable.

Table 8: Explained variance of the factor responses on the held-out training images, means over seeds 1–6 (principal-component maps: seeds 1–4). Group columns give the *cat* factor on MetaShift (cat indoors, cat outdoors, dog indoors, dog outdoors) and the *water bird* factor on Waterbirds (landbird on land, landbird on water, waterbird on land, waterbird on water).

Dataset	Features	All factors	Residuals	(0,0)	(0,1)	(1,0)	(1,1)
MetaShift	SCD student	0.22	0.00	0.47	0.07	0.10	0.47
	Shuffled-map student	0.13	0.00	0.24	0.08	0.19	0.29
	Principal-component student	0.16	0.00	0.32	0.05	0.17	0.34
	CLIP	0.74	0.15	0.92	0.88	0.83	0.91
Waterbirds	SCD student	0.26	0.00	0.29	0.14	−0.05	0.23
	Shuffled-map student	0.17	0.00	0.24	−0.09	−0.18	0.12
	Principal-component student	0.18	0.00	0.25	−0.13	−0.14	0.09
	CLIP	0.88	0.17	0.88	0.78	0.90	0.87

8 LIMITATIONS

All SCD results are validation results with the probe selected on the same split; held-out test runs with a locked probe are in progress, so the test comparison with CoSpRo remains open. The study uses one concept vocabulary per dataset, one map construction and six (two on Spur-CIFAR10) student seeds; MetaShift's seed spread (5.4 WGA points) is as large as the gain over its controls on single seeds. The configuration with the 20% hold-out scored above the same loss with every image in it (Waterbirds 49.53, MetaShift 46.70 WGA); the difference lies within the seed spread and has no explanation yet. The group-balanced probe uses labels and group annotations, so only concept discovery, map construction and SSL are label-free.

Absolute accuracies are low, since ResNet-18 trains from scratch on 1,700 to 4,795 images: SCD reaches 60 to 61% average accuracy on Waterbirds and MetaShift. LateTVG reports 55.4 and 59.3 WGA with SimCLR on Waterbirds and MetaShift under its own protocol (Hamidieh et al., 2024); a first LateTVG run under our protocol reaches 47.9 on Waterbirds (one seed) and the matched multi-seed comparison is in progress. On Waterbirds dense CLIP components match the concept maps, so the concept names add value where the relevant contrast is a weak direction of CLIP.

The method depends on CLIP's external pretraining, its vocabulary coverage and its patch resolution. Spur-CIFAR10 images are upsampled from 32 to 448 pixels for the maps, which adds no detail and CelebA, whose spurious attribute is gender, has no concept maps yet. The student keeps the training

co-occurrence: its concept scores drop in the minority groups and the spurious attribute stays linearly readable.

9 Conclusion

SCD uses dense maps of named concepts to supervise every location of a SimCLR student through a regression cross-fitted between halves of the batch, with no labels and no concept type during pretraining. On Waterbirds, MetaShift and Spur-CIFAR10 its mean validation worst-group and average accuracies exceed those of SimCLR and of the same loss on shuffled maps. Under the matched Waterbirds protocol it exceeds our earlier relational method CoSpRo and it extends the gain to MetaShift and Spur-CIFAR10, where CoSpRo stayed at the SimCLR level, while dropping CoSpRo's assumption about the spurious attribute. On MetaShift the named concepts beat dense CLIP components of the same width; on Waterbirds the two tie. The student's concepts still follow the training co-occurrence, so the results support a protocol-specific robustness gain from a stronger core feature, which locked test runs will confirm or revise.

References

M. R. Arefin, Y. Zhang, A. Baratin, F. Locatello, I. Rish, D. Liu and K. Kawaguchi. Unsupervised Concept Discovery Mitigates Spurious Correlations. *ICML*, pp. 1672–1688, 2024. https://proceedings.mlr.press/v235/arefin24a.html.

L. Bertinetto, J. F. Henriques, P. H. S. Torr and A. Vedaldi. Meta-learning with Differentiable Closed-Form Solvers. *ICLR*, 2019. https://arxiv.org/abs/1805.08136.

U. Bhalla, A. Oesterling, S. Srinivas, F. P. Calmon and H. Lakkaraju. Interpreting CLIP with Sparse Linear Concept Embeddings (SpLiCE). *NeurIPS*, 2024. https://arxiv.org/abs/2402.10376.

T. Chen, S. Kornblith, M. Norouzi and G. Hinton. A Simple Framework for Contrastive Learning of Visual Representations. *ICML*, pp. 1597–1607, 2020. https://proceedings.mlr.press/v119/chen20j.html.

V. Chernozhukov et al. Double/Debiased Machine Learning for Treatment and Structural Parameters. *The Econometrics Journal*, 21(1), 2018. https://arxiv.org/abs/1608.00060.

T. Darcet, M. Oquab, J. Mairal and P. Bojanowski. Vision Transformers Need Registers. *ICLR*, 2024. https://arxiv.org/abs/2309.16588.

A. Ghorbani, J. Wexler, J. Zou and B. Kim. Towards Automatic Concept-based Explanations. *NeurIPS*, 2019. https://arxiv.org/abs/1902.03129.

K. Hamidieh et al. Views Can Be Deceiving: Improved SSL Through Feature Space Augmentation. *ICLR*, 2024. https://arxiv.org/abs/2406.18562.

K. He, X. Zhang, S. Ren and J. Sun. Deep Residual Learning for Image Recognition. *CVPR*, 2016. https://arxiv.org/abs/1512.03385.

K. He, G. Gkioxari, P. Dollár and R. Girshick. Mask R-CNN. *ICCV*, 2017. https://arxiv.org/abs/1703.06870.

O. J. Hénaff et al. Efficient Visual Pretraining with Contrastive Detection. *ICCV*, 2021. https://arxiv.org/abs/2103.10957.

J. Hewitt and P. Liang. Designing and Interpreting Probes with Control Tasks. *EMNLP*, 2019. https://arxiv.org/abs/1909.03368.

A. E. Hoerl and R. W. Kennard. Ridge Regression: Biased Estimation for Nonorthogonal Problems. *Technometrics*, 12(1), 1970.

B. Kim, M. Wattenberg, J. Gilmer, C. Cai, J. Wexler, F. Viégas and R. Sayres. Interpretability Beyond Feature Attribution: Quantitative Testing with Concept Activation Vectors (TCAV). *ICML*, pp. 2668–2677, 2018. https://proceedings.mlr.press/v80/kim18d.html.

P. Kirichenko, P. Izmailov and A. G. Wilson. Last Layer Re-Training is Sufficient for Robustness to Spurious Correlations. *ICLR*, 2023. https://arxiv.org/abs/2204.02937.

P. W. Koh, T. Nguyen, Y. S. Tang, S. Mussmann, E. Pierson, B. Kim and P. Liang. Concept Bottleneck Models. *ICML*, pp. 5338–5348, 2020. https://proceedings.mlr.press/v119/koh20a.html.

A. Krizhevsky. Learning Multiple Layers of Features from Tiny Images. Technical report, University of Toronto, 2009.

M. Lan et al. ClearCLIP: Decomposing CLIP Representations for Dense Vision-Language Inference. *ECCV*, 2024. https://arxiv.org/abs/2407.12442.

K. Lee, S. Maji, A. Ravichandran and S. Soatto. Meta-Learning with Differentiable Convex Optimization. *CVPR*, 2019. https://arxiv.org/abs/1904.03758.

W. Liang and J. Zou. MetaShift: A Dataset of Datasets for Evaluating Contextual Distribution Shifts and Training Conflicts. *ICLR*, 2022. https://arxiv.org/abs/2202.06523.

E. Z. Liu, B. Haghgoo, A. S. Chen, A. Raghunathan, P. W. Koh, S. Sagawa, P. Liang and C. Finn. Just Train Twice: Improving Group Robustness without Training Group Information. *ICML*, pp. 6781–6792, 2021. https://proceedings.mlr.press/v139/liu21f.html.

W. Luo, Y. Li, R. Urtasun and R. Zemel. Understanding the Effective Receptive Field in Deep Convolutional Neural Networks. *NeurIPS*, 2016. https://arxiv.org/abs/1701.04128.

T. Oikarinen, S. Das, L. M. Nguyen and T.-W. Weng. Label-Free Concept Bottleneck Models. *ICLR*, 2023. https://arxiv.org/abs/2304.06129.

A. van den Oord, Y. Li and O. Vinyals. Representation Learning with Contrastive Predictive Coding. *arXiv:1807.03748*, 2018. https://arxiv.org/abs/1807.03748.

S. Petryk et al. On Guiding Visual Attention with Language Specification. *CVPR*, 2022. https://arxiv.org/abs/2202.08926.

A. Radford et al. Learning Transferable Visual Models From Natural Language Supervision. *ICML*, pp. 8748–8763, 2021. https://proceedings.mlr.press/v139/radford21a.html.

M. Ranzinger et al. AM-RADIO: Agglomerative Vision Foundation Model – Reduce All Domains Into One. *CVPR*, 2024. https://arxiv.org/abs/2312.06709.

J. Robinson, L. Sun, K. Yu, K. Batmanghelich, S. Jegelka and S. Sra. Can contrastive learning avoid shortcut solutions? *NeurIPS*, pp. 4974–4986, 2021. https://arxiv.org/abs/2106.11230.

A. S. Ross, M. C. Hughes and F. Doshi-Velez. Right for the Right Reasons: Training Differentiable Models by Constraining their Explanations. *IJCAI*, 2017. https://arxiv.org/abs/1703.03717.

C. K. Ryali, D. J. Schwab and A. S. Morcos. Characterizing and Improving the Robustness of Self-Supervised Learning through Background Augmentations. *arXiv:2103.12719*, 2021. https://arxiv.org/abs/2103.12719.

S. Sagawa, P. W. Koh, T. B. Hashimoto and P. Liang. Distributionally Robust Neural Networks for Group Shifts: On the Importance of Regularization for Worst-Case Generalization. *ICLR*, 2020. https://arxiv.org/abs/1911.08731.

F. Träuble et al. On Disentangled Representations Learned from Correlated Data. *ICML*, 2021. https://arxiv.org/abs/2006.07886.

J. von Kügelgen, Y. Sharma, L. Gresele, W. Brendel, B. Schölkopf, M. Besserve and F. Locatello. Self-Supervised Learning with Data Augmentations Provably Isolates Content from Style. *NeurIPS*, 2021. https://arxiv.org/abs/2106.04619.

F. Wang, J. Mei and A. Yuille. SCLIP: Rethinking Self-Attention for Dense Vision-Language Inference. *ECCV*, 2024. https://arxiv.org/abs/2312.01597.

X. Wang et al. Dense Contrastive Learning for Self-Supervised Visual Pre-Training. *CVPR*, 2021. https://arxiv.org/abs/2011.09157.

Z. Xie et al. Propagate Yourself: Exploring Pixel-Level Consistency for Unsupervised Visual Representation Learning. *CVPR*, 2021. https://arxiv.org/abs/2011.10043.

C. Yang et al. CLIP-KD: An Empirical Study of CLIP Model Distillation. *CVPR*, 2024. https://arxiv.org/abs/2307.12732.

C. Zhang, S. Bengio, M. Hardt, B. Recht and O. Vinyals. Understanding Deep Learning Requires Rethinking Generalization. *ICLR*, 2017. https://arxiv.org/abs/1611.03530.

B. Zhou, A. Khosla, A. Lapedriza, A. Oliva and A. Torralba. Learning Deep Features for Discriminative Localization. *CVPR*, 2016. https://arxiv.org/abs/1512.04150.

C. Zhou, C. C. Loy and B. Dai. Extract Free Dense Labels from CLIP. *ECCV*, 2022. https://arxiv.org/abs/2112.01071.

W. Zhu, S. Liu, C. Fernandez-Granda and N. Razavian. Making Self-supervised Learning Robust to Spurious Correlation via Learning-speed Aware Sampling. *arXiv:2311.16361*, 2023. https://arxiv.org/abs/2311.16361v2.

A CONFIGURATION

Component	Setting
CLIP teacher	OpenCLIP ViT-B/32 with LAION-2B weights
SpLiCE	L1 penalty 0.25 on whole-image embeddings; LAION 10k vocabulary (MetaShift), Open Images v7 vocabulary (Waterbirds, Spur-CIFAR10)
Concept groups	Average linkage on raw text embeddings, text cosine 0.85; response correlation at least 0.5; words active on $[0.002, 0.95]$ of images
Concept factors	Groups active on $[0.02, 0.90]$ of images: 242 (Waterbirds), 130 (MetaShift), 147 (Spur-CIFAR10)
Concept maps	Bicubic resize to 448×448; nine 224-pixel windows at offsets $\{0, 112, 224\}$; ClearCLIP last block; 16-pixel cells averaged over windows; average-pooled to 14×14; softmax over factors at $\tau_M = 0.01$; float16
Targets	Standardized per factor over all images and cells; RoIAlign with 2×2 samples per bin over the view's crop box; mirrored for flipped views
Student features	Last residual stage of ResNet-18: $512 \times 7 \times 7$ (224-pixel views), $512 \times 4 \times 4$ (Spur-CIFAR10, 32-pixel views)
Cross-fit	First and second half of the batch's images, both views of an image in one half; unit-norm rows; absolute ridge $\rho = 1$; float32 solve
Loss weight	0 up to epoch 10, linear warm-up to $\lambda_{\max} = 5$ by epoch 20 (chosen from 1, 3, 5 and 10)
Hold-out	20% of training images with loss weight 0, the same images in every method and seed
SSL	SimCLR, 500 epochs, batch 128, SGD with learning rate 0.01 decayed by 0.1 at epochs 350, 400 and 450, weight decay 10^{-4}, temperature 0.05, float16 autocast, crop scale $[0.2, 1]$
Probe	Logistic regression with L2 coefficient 0.001 every 25 epochs on a group-balanced training subset (224 / 400 / 1,400 images)

B PER-SEED AND PER-GROUP RESULTS

Table 9: Waterbirds validation. Avg. and WGA: mean of the last four probes; group columns: probe at epoch 500. Groups (y, a): landbird on land, landbird on water, waterbird on land and waterbird on water. Values are percentages.

Method	Seed	Avg.	WGA	(0,0)	(0,1)	(1,0)	(1,1)
SimCLR	1	50.67	44.97	44.54	50.00	54.89	66.92
SimCLR	2	52.00	43.79	43.68	58.37	52.63	57.89
SimCLR	3	52.90	47.43	47.32	56.22	51.88	59.40
SimCLR	4	53.44	46.99	53.75	50.43	46.62	70.68
SimCLR	5	52.15	48.18	47.54	50.86	58.65	63.16
SimCLR	6	53.48	48.61	48.61	55.58	57.89	60.15
Shuffled maps	1	56.46	50.19	53.32	60.52	50.38	57.89
Shuffled maps	2	53.29	47.54	47.75	56.87	59.40	54.14
Shuffled maps	3	52.61	44.91	45.61	59.44	51.88	55.64
Shuffled maps	4	48.02	44.27	43.90	45.28	55.64	60.15
Shuffled maps	5	53.19	48.07	48.18	56.22	54.14	58.65
Shuffled maps	6	53.17	50.00	50.75	56.22	51.13	54.14
SCD	1	59.97	57.26	58.03	58.80	57.89	74.44
SCD	2	63.45	51.13	58.24	72.96	51.13	60.90
SCD	3	59.70	50.75	50.54	63.73	71.43	68.42
SCD	4	57.49	52.96	52.68	60.09	54.89	68.42
SCD	5	63.55	58.65	58.67	70.60	59.40	58.65
SCD	6	57.78	45.82	45.61	64.38	64.66	73.68

On Spur-CIFAR10, SCD reaches $70.03/37.77$ and $68.61/38.49$ Avg./WGA on seeds 1 and 2, SimCLR $59.39/16.36$ and $59.71/14.89$ and the shuffled maps $56.41/15.84$ and $55.39/18.65$. Its 100 groups pair each class with each line colour.

Table 10: MetaShift validation. Avg. and WGA: mean of the last four probes; group columns: probe at epoch 500. Groups (y, a): cat indoors, cat outdoors, dog indoors and dog outdoors. Values are percentages.

Method	Seed	Avg.	WGA	(0,0)	(0,1)	(1,0)	(1,1)
SimCLR	1	56.34	50.69	59.72	50.00	58.33	55.56
SimCLR	2	55.30	44.79	50.00	62.50	44.44	63.89
SimCLR	3	48.09	31.60	48.61	30.56	50.00	61.11
SimCLR	4	59.46	52.08	61.11	51.39	58.33	66.67
SimCLR	5	53.82	44.44	62.50	45.83	44.44	63.89
SimCLR	6	57.99	42.36	58.33	41.67	61.11	70.83
Shuffled maps	1	58.07	43.06	69.44	45.83	43.06	72.22
Shuffled maps	2	54.25	42.01	48.61	43.06	52.78	73.61
Shuffled maps	3	56.94	49.31	62.50	48.61	50.00	65.28
Shuffled maps	4	53.21	46.87	62.50	47.22	47.22	52.78
Shuffled maps	5	58.85	46.53	62.50	45.83	48.61	77.78
Shuffled maps	6	55.12	42.71	43.06	43.06	55.56	77.78
SCD	1	66.06	60.42	69.44	61.11	68.06	66.67
SCD	2	58.85	50.69	59.72	51.39	51.39	75.00
SCD	3	61.63	54.17	62.50	52.78	58.33	70.83
SCD	4	62.33	46.87	69.44	47.22	55.56	79.17
SCD	5	60.33	48.61	63.89	48.61	56.94	73.61
SCD	6	57.73	45.83	58.33	45.83	52.78	72.22

Table 11: Paired SCD differences in percentage points, mean of the last four probes.

	Waterbirds				MetaShift			
	vs. SimCLR		vs. shuffled maps		vs. SimCLR		vs. shuffled maps	
Seed	ΔAvg.	ΔWGA	ΔAvg.	ΔWGA	ΔAvg.	ΔWGA	ΔAvg.	ΔWGA
1	+9.30	+12.29	+3.50	+7.07	+9.72	+9.72	+7.99	+17.36
2	+11.45	+7.34	+10.15	+3.59	+3.56	+5.90	+4.60	+8.68
3	+6.80	+3.32	+7.09	+5.84	+13.54	+22.57	+4.69	+4.86
4	+4.05	+5.97	+9.47	+8.69	+2.86	−5.21	+9.11	+0.00
5	+11.41	+10.47	+10.36	+10.57	+6.51	+4.17	+1.48	+2.08
6	+4.30	−2.78	+4.61	−4.18	−0.26	+3.47	+2.60	+3.13
Mean	+7.88	+6.10	+7.53	+5.26	+5.99	+6.77	+5.08	+6.02